AF-1984 MIT 2E6

M.Sc DEGREE EXAMINATION, APRIL 2010

Second Semester

INFORMATION TECHNOLOGY

ELECTIVE - DIGITAL IMAGE PROCESSING

(CBCS - 2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 x 2 = 20)

Answer **ALL** Questions

1. Define Image Processing.
2. Write about image formation in the eye.
3. Define Bitplane slicing.
4. Distinguish low pass filter and high pass filter.
5. Write the use of restoration.
6. Define inverse filter.
7. What is known as interpixel redundancy?
8. Distinguish lossless and lossy compression.
9. Write any two applications of pattern recognition..
10. Mention any two basic problems in pattern recognition system.

Part-B (5 x 5 = 25)

Answer ALL Questions

11. a. Explain DFT.

(OR)

b. Describe Walsh Transform in detail.

12. a. Write in detail about Homomorphic filtering.

(OR)

b. Explain gray level slicing and bit plane slicing.

13. a. Discuss briefly about Algebraic approach to restoration.

(OR)

b. Write in detail about least mean square filter.

14. a. Explain fidelity criteria in detail.

(OR)

b. Discuss lossless predictive coding in detail.

15. a. Write short notes on Ho Kashyap procedure.

(OR)

b. Explain AI approach to pattern recognition problem.

Part-C (3 x 10 = 30)

Answer any THREE Questions

16. Explain elements of Digital Image Processing System with a neat diagram.

17. Discuss Histogram processing in detail.

18. Describe in detail about diagonalization of circulant and block circulant matrices.

19. Discuss in detail about lossless compression.

20. Explain Linear discriminant function pattern classification using statistical approach in detail.

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AF-1971 **MIT1C2**

M.Sc. DEGREE EXAMINATION, APRIL 2010

First Semester

Information Technology

C AND DATA STRUCTURE

(CBCS—2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **All** questions.

1. What are identifiers ?

2. Write about declaring variables in C.

3. Write the general format of the if.. else statement.

4. What is the necessity of for statement in C ?

5. What is a two dimensional array ?

6. What are strings ?

7. What is meant by Merging lists ?

8. Write any two applications of Doubly Linked List.

9. Write the types of Binary Tree.

10. What is Binary search ?

Part - B (5 × 5 = 25)

Answer **All** questions.

11. (a) Write the rules for defining the variables.

(*Or*)

(b) Write in detail about Input and output in C.

12. (a) Discuss on GoTo and Continue statement.

(*Or*)

(b) Write brief notes on do... while statement.

13. (a) Explain about predefined streams.

(*Or*)

(b) Discuss about Arithmetic operations with pointers.

14. (a) Discuss about Traversal of a List.

(*Or*)

(b) Write short notes on Representation of stack.

15. (a) Write brief notes on Binary search Tree.

(*Or*)

(b) Write in detail about Tree sort.

Part - C

(3 × 10 = 30)

Answer any **Three** questions.

16. Explain about Basics of C and the Library Function in C.

17. Write brief notes on :

    (a) Nested swith ( ) case statement.

    (b) Nested For statement.

18. Explain about working with strings and standard functions.

19. Discuss on Doubly Linked List in detail.

20. Explain in detail about Insertion sort.

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AF-1972 MIT1C3

M.Sc. DEGREE EXAMINATION, APRIL 2010

Information Technology

OBJECT ORIENTED PROGRAMMING IN JAVA

(CBCS—2008 Onwards)

Duration : 3 Hours Maximum : 75 marks

Part - A (10 × 2 = 20)

Answer **All** questions.

1. Define Hot Java.

2. Write the syntax of do ....while statement.

3. What is meant by exception handling ?

4. Define Methods.

5. What do you mean by synchronization ?

6. Define thread priority.

7. List out any two difference between applet and application programs.

8. List out the classes in the java lang package.

9. What are the JDBC products ?

10. Define JSP.

Part - B (5 × 5 = 25)

Answer **All** questions.

11. (a) Explain the processes of developing and executing java program.

Or

(b) What is constant ? How it is classified ?

12. (a) Writer a java program to Find sum of odd number in the range of 1 to N

Or

(b) Writer a java program to Arranging the numbers in Ascending Order.

13. (a) How to Block and Stop the thread ? Explain.

Or

(b) Write about the Multi-threaded program.

14. (a) Write about the local and Remote applet.

Or

(b) Write a applet to display the current date.

15. (a) Explain the JDBC-ODBC bridge.

Or

(b) Explain the JDBC-Net All - Java driver.

Part -C (3 × 10 = 30)

*Answer any **Three** questions.*

16. Write a java program for matrix multiplication.

17. Discuss about the Input / Output classes.

18. Explain the life-cycle of an thread.

19. Describe a life cycle of an applet.

20. Explain the various steps to connect a JDBC driver.

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AF-1973 **MIT1E1**

M.Sc. DEGREE EXAMINATION, APRIL 2010

First Semester

Information Technology

PROBABILITY AND STATISTICS

(CBCS—2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **All** the questions

1. Two events A and B are mutually exclusive if $P(A \cap B) =$ ..... and are independent if $P(A \cap B) =$ .....

2. Two six faced dice are thrown. Find the probability that the product is 2.

3. Check whether f(x) = x, x = $\frac{1}{16}, \frac{3}{16}, \frac{1}{4}, \frac{1}{2}$ is a probability mass function.

4. Define the mathematical expection of a random variable.

5. Find the mean of the Binomial distribution.

$$\binom{10}{x}\left(\frac{2}{3}\right)^{3}\left(\frac{3}{5}\right)^{10-x}, x = 0, 1, 2 \ldots 10.$$

6. The two lines of regression are given as $x + 2y - 5 = 0$ and $2x + 3y = 8$. Find the mean values of x and y.

7. What do you mean by standard error ?

8. What is Chi-square test of goodness of fit ?

9. What is randomised block design ?

10. What is the assumption made in the analysis of Latin square design ?

Part - B $(5 \times 5 = 25)$

Answer **All** the questions

11 a. A bag contains 5 white and 3 black balls. Two balls are drawn one after the other without replacement. Find the probability that second ball drawn is black given that the first ball drawn in white.

(*Or*)

b. A town has two doctors X and Y operating independently. If the probability that doctor X is available in 0.9 and that for Y is 0.8. what is the probability that at least one doctor is available when needed ?

12 a. Given the following table :

x :	−3	−2	−1	0	1	2	3
p(x) :	0.05	0.10	0.30	0	0.30	0.15	0.10

Compute E(X) and variance of X.

(*Or*)

b. Find the moment generating function of random variable X having the p.d.f.

$$f(x) = \begin{cases} 1/3, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}.$$

13 a. If X has exponential distribution with mean 2, find $p(X < 1 \mid X < 2)$

(*Or*)

b. The following data pertain to the marks in subject A and B in an examinatin.

Mean marks in A = 39.5, Mean marks in B = 47.5, Standard deviation of marks in A = 10.8, Standard deviation of marks in B = 16.8. Coefficient of correlation between marks in A and marks in B = 0.42. Find the two lines of regression.

14 a. A sample of heights of 6400 soldiers has a mean of 67.85 inches and a S.D. of 2.56 inches while another sample of heights of 1600 sailors has a mean of 68.55 inches with a S.D. of 2.52 inches. Do the data indicate that the sailor are on the average taller that soldiers ?

(*Or*)

b. The 9 items of a sample have the following values : 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these values differ significantly from the assumed mean 47.5 ? (For $\nu = 8$, $t_{0.05} = 2.31$).

15 a. Production of rice in a district during the last 10 years is given below. Determine the trend using 4 yearly moving average.

Year :	1960	1961	1961	1963	1964	1965	1966	1967	1968	1969
Production : (in '000 tons)	112	123	106	134	138	145	116	143	136	154

(*Or*)

b. Fit a straight line trend to the following data by the method of least squares :-

Year :	1971	1972	1973	1974	1975	1976	1977	1978	1979
Production :	1.3	2.4	4.3	2.4	3.3	5.6	4.0	4.7	5.3

Part - C

$(3 \times 10 = 30)$

Answer any **Three** questions

16. Three machines X, Y and Z of equal capacities are producing bullets. The probabilities that machines produce defective (bullets which do not confirm to specifications) are 0.1, 0.2 and 0.1 respectively. A bullet is taken at random for a day's production and is found to be defective. What is the probability that it came from machine X ?

17. The joint probability distribution of two random variables x and y is given by : $p(X = 0, Y = 1) = {}^1/_3$, $p(X = 1, Y = -1) = {}^1/_3$, $p(X = 1, Y = 1) = {}^1/_3$. Find (i) marginal distributions of X and Y. (ii) the conditional probability distribution of X given Y = 1.

18. Calculate the coefficient of correlation for the following :-

X:	65	66	67	67	68	69	70	72
Y:	67	68	65	68	72	72	69	71

19. A die is thrown 60 times with the following results :-

Face	:	1	2	3	4	5	6
Frequency	:	8	7	12	8	14	11

Test at 5% level of significance if one die is unbiased. (Given $\chi^2_{0.05}$ for 5 d.f is 11.1)

20. Three varieties A, B, C of a crop are tested in a randomized block design with four replications. The plot yields in pounds are as follows:

A6	C5	A8	B9
C8	A4	B6	C9
B7	B6	C10	A6

Analyse the experimental yield and state your conclusion.

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AF-1974 MIT1E2

M.Sc. DEGREE EXAMINATION, APRIL 2010

First Semester

Information Technology

Elective—OBJECT ORIENTED ANALYSIS AND DESIGN

(CBCS—2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **All** questions.

1. Define meta data.

2. What are the various types of association ?

3. What is an event ?

4. Define a Data Flow.

5. Define object modeling.

6. What is meant by reusability ?

7. What is a Sub system ?

8. Define ATM network.

9. State (any 2) some hints on designing algorithms.

10. How do you design the association in object Design.

Part - B

$(5 \times 5 = 25)$

Answer **All** questions.

11. (a) Explain about multiple inheritance with an example.

Or

(b) Justify : Objects and Classes are reusable.

12. (a) How do you represent functional models ? Discuss.

Or

(b) Justify : Data flow diagram represents functional model.

13. (a) Explain about Dynamic modeling in detail.

Or

(b) How do you add operations in functional modeling ?

14. (a) Explain about Handling boundary condition in detail.

Or

(b) Explain about identifying concurrency in detail.

15. (a) Explain the issues involved in packaging.

Or

(b) How do you increase the amount of inheritance.

Part -C

$(3 \times 10 = 30)$

Answer any **Three** questions.

16. What is Aggregation ? Explain in detail.

17. Discuss and give some concepts on relation of objects and dynamic constraints.

18. Explain about object oriented analysis process in detail.

19. Explain the Architecture of ATM system with Block diagram.

20. How the analysis and Design react in physical packaging of object and how we make documentation on that object ?

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AF-1975

MIT1E3

M.Sc. DEGREE EXAMINATION, APRIL 2010

First Semester

Information Technology

Elective—OPTIMIZATION TECHNIQUES

(CBCS—2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **All** the questions.

1. What is meant by degeneracy of a CPP ?

2. What is infeasible solution ? How it is identified in the simplex table ?

3. What is a dual problem in linear programming ?

4. Differentiate between Pure and Mixed integer programming problems.

5. Define critical path of a project Network.

6. What is a dummy activity ?

7. Explain Kend all's notation of Queues.

8. Write Little's formula.

9. What is meant by Inventory ?

10. Define lead time.

Part - B $(5 \times 5 = 25)$

Answer **All** the questions.

11. (a) A firm produce three types of Cloth, say A,B,C. Three kinds of wool are required for it, say, red wool, green wool, and blue wool. One unit length of type A cloth needs 2 yards of red wool and 3 yards of blue wool, one unit length of type B cloth needs 3 yards of red wool, 2 yards of green wool and 2 yards of the blue wool and one unit length of type C cloth needs 5 yards of green wool and 4 yards of blue wool. The firm has stock of only 8 yards of red wool, 10 yards of green wool and 15 yards of blue wool. It is assumed that the income obtained from one unit length of type A cloth is Rs. 30 of type B cloth is Rs 50 and that of type C cloth is Rs. 40 Formulate the problem as linear programming problem.

(*Or*)

(b) Solve graphically

Maximize : $2 = 3x_1 + 2x_2$

subject to $2x_1 + x_2 \leq 10$

$x_1 + 3x_2 \leq 6$

$x_1, x_2 \geq 0.$

12. (a) Describe Gomory's integer programming problem and its algorithm.

(*Or*)

(b) Explain the Branch and Bound method in integer programming problem.

13. (a) Draw the network for the project whose activities with their predecessor relationships are given below :—

A, C, D can start simultaneously E > B, C; G > D; H, I > E, F; J > I, G ; K > H; B > A.

(*Or*)

(b) What are the basic differences between PERT and CPM ?

14. (a) Describe the Queueing model M/M/1.

(*Or*)

(b) In a Public Telephone booth the arrivals are on the average 15 per hour. A call on the average takes 3 minutes. If there is just one phone find expected number of callers in booth at any time.

15. (a) The annual demand for an item is 3200 units. The unit cost is Rs 6/= and inventory carrying charges 25% per annum. If the cost of one procurement is Rs 150/= determine Economic order quantity.

(*Or*)

(b) Explain ABC selective control techniques.

Part - C

$(3 \times 10 = 30)$

Answer any **three** the questions.

16. Solve by simplex method :—

Minimize $z = x_1 - 3x_2 + 2x_3$

subject to
$$3x_1 - x_2 + 2x_3 \geq 7$$
$$-2x_1 + 4x_2 \leq 12$$
$$-4x_1 + 3x_2 + 8x_3 \leq 10$$
$$x_1, x_2, x_3 \geq 0$$

17. Solve the following IPP :

Maximize $z = 5x_1 + 7x_2$

subject to
$$-2x_1 + 3x_2 \leq 6$$
$$6x_1 + x_2 \leq 30$$
$$x_1, x_2 \geq 0$$

and integers.

18. Draw the network and find the critical path, also find various floats for the following :—

Activity	:	0—1	1—2	1—3	2—4	2—5
Duration	:	3	8	12	6	3

Activity	:	3—4	3—6	4—7	5—7	6—7
Duration	:	3	8	5	3	8

19. Cars arrive at a petrol pumb, having one petrol unit, in Poisson fashion with an average of 10 cars per hour. The service time is distributed exponentially with a mean of 3 minutes. Find (i) average number of cars in the system. (ii) average waiting time in the Queue. (iii) average Queue length.

20. The demand for an item in a company is 18,000 units per year, and the company can produce the item at a rate of 3,000 per month. The cost of one set up is Rs. 500 and the holding cost of one unit per month is 15 paise. The shortage cost of one unit is Rs. 20 per month. Determine the optimum manufacturing quantity omcl the number of shortages. Also determine the manufacturing time and time between set-ups.

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AF-1976 **MIT 2C1**

M.Sc. DEGREE EXAMINATION, APRIL 2010

Second Semester

INFORMATION TECHNOLOGY

COMPUTER NETWORKS

(CBCS - 2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 ×2 = 20)

Answer **All** Questions.

1. Define a Computer Network.

2. List the top most layer in OSI reference model.

3. What is Analog Transmission?

4. List any two Switching Techniques.

5. What is a MAN?

6. What is ALOHA?

7. List any two Routing algorithms.

8. What is congestion?

9. What is Encryption?

10. Expand FTP.

Part - B (5 × 5 = 25)

Answer **All** Questions.

11. a. Explain briefly the Applications of Computer Networks.

(*Or*)

b. Write a brief note on Bus type LAN structure.

12. a. Write a brief note on Digital Transmission.

(*Or*)

b. Explain briefly about Terminal handling.

13. a. Explain briefly about Data Link layer Design issues.

(*Or*)

b. Write a brief note on Error detection.

14. a. Explain briefly about session layer design issues.

(*Or*)

b. What is Remote Procedure call? Briefly explain.

15. a. Explain briefly about Abstract Syntax Notation.

(*Or*)

b. Write a brief note on Electronic mail.

Part - C (3 × 10 = 30)

Answer any **Three** Questions.

16. Describe in detail the ISO OSI Reference model in detail.

17. Explain in detail about Transmission and Switching.

18. Describe in detail the HDLC Protocol.

19. Explain in detail the congestion control algorithms.

20. Explain in detail about Data Compression Techniques.

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AF-1977 MIT 2C2

M.Sc DEGREE EXAMINATION, APRIL 2010

Second Semester

INFORMATION TECHNOLOGY

RDBMS

(CBCS - 2008 Onwards)

Duration : 3 Hours Maximum : 75 marks

Part - A (10 x 2 = 20)

Answer **ALL** Questions

1. Write down the functional components of E-R diagram.

2. Define the term hashing.

3. What is an index?

4. Define B-Tree.

5. List down the fundamental operations in Relational Algebra.

6. What do you mean by normal forms?

7. Define Security.

8. Define Cryptosystem.

9. Mention some DML commands.

10. Write the syntax for Privilege statement.

Part-B (5 x 5 = 25)

Answer **ALL** Questions

11. a. Discuss on components of DBMS.

(OR)

b. Explain about Entity Relationship.

12. a. Explain deferred splitting in detail.

(OR)

b. Write about B-Tree Indexed file.

13. a. Discuss on Relational Calculus with examples.

(OR)

b. Explain any two normal forms with suitable example.

14. a. Discuss on statistical database.

(OR)

b. Explain about Database Recovery.

15. a. Discuss Embedded SQL.

(OR)

b. Explain the following with example:

i. Modifying Table

ii. Renaming Table

Part-C (3 x 10 = 30)

Answer any **THREE** Questions

16. Brief about file organization.

17. Explain file organization based on dynamic hashing.

18. Discuss Relational Algebra with example.

19. Explain about Database System Architecture.

20. Write a brief account on SQL.

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AF-1978

MIT 2C3

M.Sc DEGREE EXAMINATION, APRIL 2010

Second Semester

INFORMATION TECHNOLOGY

OPERATING SYSTEM

(CBCS - 2008 Onwards)

Duration : 3 Hours

Maximum : 75 marks

Part - A (10 x 2 = 20)

Answer **ALL** Questions

1. What is batch system?
2. Define virtual machine.
3. What are kernel threads?
4. Define cooperating process.
5. Write any two advantages of process synchronization.
6. What are semaphores?
7. Define swapping.
8. Give the purpose of paging.
9. Give the purpose of file management.
10. What is tertiary storage?

Part-B (5 x 5 = 25)

Answer ALL Questions

11. a. Write the advantages of real time system.

(OR)

b. Justify about system calls.

12. a. Explain about process states.

(OR)

b. Illustrate scheduling criteria.

13. a. Discuss in detail about two task solutions.

(OR)

b. Discuss about thread scheduling.

14. a. Write short notes on thrashing.

(OR)

b. What are different page replacement algorithms? Explain.

15. a. Discuss in detail about directory structure.

(OR)

b. Describe about disk reliability.

Part-C (3 x 10 = 30)

Answer any THREE Questions

16. Write short notes on the following:

    i. Parallel system

    ii. Distributed system

17. Explain in detail about CPU scheduling algorithms.

18. Describe about deadlock algorithms.

19. Discuss about demand paging.

20. Write an overview about disk management.

——— XXX ———

AF-1979 MIT 2E1

M.Sc DEGREE EXAMINATION, APRIL 2010

II Semester

INFORMATION TECHNOLOGY

ELECTIVE - MOBILE COMPUTING

(CBCS - 2008 Onwards)

Duration : 3 Hours Maximum : 75 marks

Part - A (10 x 2 = 20)

Answer ALL Questions

1. What is IP routing?
2. What is IETF?
3. Define modulation.
4. What is GSM?
5. What is Agent?
6. Define Authentication.
7. What is meant by Tunnel?
8. What is encapsulation?
9. State the notation for IPV6.
10. What is WAP?

Part-B (5 x 5 = 25)

Answer ALL Questions

11. a. Illustrate and explain IEEE 802.116 protocol architecture.

(OR)

b. Explain entities and terminologies ‘mobile nodes’ and ‘home agent’.

12. a. Explain any two multiplexing mechanisms.

(OR)

b. Explain direct sequence spread spectrum.

13. a. Illustrate and explain agent advertisement messages.

(OR)

b. Explain the mobile node registration process with home agent.

14. a. Illustrate and explain minimal encapsulation.

(OR)

b. Explain agent discovery mechanism.

15. a. Explain the basic architecture of cellular IP.

(OR)

b. Explain movement detection of mobile nodes.

Part-C (3 x 10 = 30)

Answer any THREE Questions

16. Explain Dynamic Source routing in detail.

17. Illustrate and explain GSM architecture.

18. Illustrate and explain mobile node authentication.

19. Explain in detail mobile route optimization.

20. Discuss mobility support in IPV6.

——— XXX ———

AF-1980 MIT 2E2

M.Sc DEGREE EXAMINATION, APRIL 2010

Second Semester

Information Technology

DATA MINING AND DATA WAREHOUSING

(CBCS— 2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **All** questions

1. What is KDD?

2. What is production factor?

3. What is enrichment?

4. List the different data mining techniques.

5. What is the use of meta data?

6. What is the need to partition the data?

7. What is data warehousing?

8. What is meant by data security?

9. Define capacity planning.

10. What is tuning?

Part-B (5 × 5 = 25)

Answer **All** questions

11. a. Write the practical applications of data mining.

(Or)

b. Describe the rules to be followed in designing the data war house.

12. a. Explain about association rules with suitable examples.

(Or)

b. Discuss about data selection and data cleaning procedures.

13. a. Describe the role of warehouse process manager.

(Or)

b. Explain different partitioning stratergies with suitable examples.

14. a. Explain the physical layout of data warehouse.

(Or)

b. Discuss about backup and recovery system.

15. a. Explain about capacity planning.

(Or)

b. Describe the tuning of data warehouse.

Part-C (3 ×10 = 30)

Answer any **Three** questions

16. Explain concept learning in detail.

17. Explain about genetic algorithms in KDD environment.

18. Describe the data warehouse process architecture.

19. Explain the hardware architecture of data warehouse.

20. Explain the features of data warehouse.

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AF-1981 **MIT 2E3**

M.Sc. DEGREE EXAMINATION, APRIL 2010

Second Semester

Information Technology

Elective - COMPUTER GRAPHICS

(CBCS - 2008 Onwards)

Duration : 3 Hours Maximum : 75 marks

Part - A (10 × 2 = 20)

Answer **All** questions.

1. What is computer animation?

2. What is graphics user interface?

3. Define intensity.

4. What is transformation?

5. What is a segment?

6. Name any two physical input devices.

7. Define scaling.

8. Write any two transformation commands.

9. Define projection.

10. What is viewing transformation?

Part - B (5 × 5 = 25)

Answer **All** questions.

11. a. Write a note on image processing.

(*Or*)

b. What is display processor?

12. a. Discuss: Bundled attributes.

(*Or*)

b. What is composite transformation? Explain.

13. a. What is window to viewpoint transformation?

(*Or*)

b. Write a note on interactive picture construction technique.

14. a. Write a brief discussion on three dimensional display techniques.

(*Or*)

b. Describe: Three dimensional graphics packages.

15. a. Write a note on projections.

(*Or*)

b. Explain: Scan line method.

Part - C (3 × 10 = 30)

Answer any **Three** questions.

16. Discuss Bresenham's line drawing algorithm with suitable examples.

17. Explain the types of transformations.

18. Write notes on the following:

i. Multiple workstations

ii. Windowing concepts

19. Discuss rotating a point about an arbitrary axis with an example.

20. Discuss: Z-buffer and A-buffer algorithms.

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AF-1982 **MIT2E4**

M.Sc DEGREE EXAMINATION, APRIL 2010

Second Semester

Information Technology

GRID COMPUTING

(CBCS —2008 Onwards)

Duration : 3 Hours Maximum : 75 marks

Part - A (10 x 2 = 20)

Answer **ALL** Questions

1. Name few Grid Applications.

2. What is Cluster Computing?

3. What is CORBA?

4. Define Semantic Grid.

5. Name the software used to enable Grid.

6. What is OGSA webservice?

7. What is proxy credential?

8. Define GRAM.

9. What is meant by GFTP?

10. What is GIS?

Part-B (5 x 5 = 25)

Answer **ALL** Questions

11. a. Explain High-Performance computing.

(OR)

b. Write a note on Grid Programming model.

12. a. Explain the Grid architecture relationship with microsofts' COM technology.

(OR)

b. Explain the concept of virtual organizations.

13. a. Explain Single System Environment cluster.

(OR)

b. How to manage data grids? Explain.

14. a. Explain messaging and queuing in OGSI.

(OR)

b. Explain metering and accounting in OGSI.

15. a. Explain in brief about OGSA platform model.

(OR)

b. Explain Data Access and Integration Service (DAIS).

Part-C (3 x 10 = 30)

Answer any **THREE** Questions

16. Discuss the evolution of Grid in detail.

17. Illustrate and explain Grid Computing Architecture model in detail.

18. Explain with neat diagram the Grid Security Infrastructure in action.

19. Illustrate and explain how a grid information service builds the bridge between information about resources on the grid and the user community.

20. Discuss the case study on molecular modeling for Drug Design.

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AF-1983 MIT 2E5

M.Sc DEGREE EXAMINATION, APRIL 2010

Second Semester

INFORMATION TECHNOLOGY

ELECTIVE - PARALLEL PROCESSING

(CBCS - 2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 x 2 = 20)

Answer **ALL** Questions

1. What is efficiency of a parallel algorithm?

2. Define control parallelism.

3. What is a cluster?

4. Differentiate linear and ring networks.

5. List down message passing mechanisms.

6. Define mapping granularity.

7. List down any two design issues of parallel algorithm.

8. Specify the complexities of parallel algorithms.

9. Differentiate shared bus and cross bar switches.

10. What is meant by memory contention?

Part-B (5 x 5 = 25)

Answer ALL Questions

11. a. Explain data parallelism with an example.

(OR)

b. Explain in brief about the issues of parallel processing.

12. a. Differentiate tightly coupled and loosely coupled systems.

(OR)

b. Explain in brief about the links in shuffle exchange networks.

13. a. Explain in brief about the precedence graph of a process.

(OR)

b. With suitable example, explain temporal parallelism.

14. a. Explain in brief about anomalies in parallel algorithms.

(OR)

b. Explain any one parallel search algorithm with an example.

15. a. Explain the working principle of multiport memory.

(OR)

b. What is meant by Cache coherence? Explain in brief.

Part-C (3 x 10 = 30)

Answer any THREE Questions

16. Discuss about the demands and mechanisms of implementing parallel processing.

17. With suitable diagrams, explain the hypercube network.

18. Differentiate message passing and shared address space mechanisms.

19. Explain detail, how the performance of parallel algorithms is analyzed.

20. With suitable examples, explain how shared variables are handled.

——— XXX ———

M.Sc. DEGREE EXAMINATION, APRIL 2010

First Semester

Information Technology

APPLIED MATHEMATICS FOR COMPUTER SCIENCE

(CBCS—2008 Onwards)

Duration : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** Questions.

1. Form the conjunction and the disjunction of *p* and *q*.

   P : I am rich, q : I am happy.

2. Show that Q => P → Q.

3. Find the principal disjunctive normal form of P ↔ Q.

4. Define rule P and rule CP in the inference theory.

5. Define connected Graph.

6. How many edges are there in a graph with 10 vertics each of degree 5 ?

7. What is meant by degeneracy in LPP.

8. Explain briefly the term 'Artificial' variables.

9. Define degenerate basic feasible solution of a transportation problem.

10. Define prohibited assignment.

Part - B $(5 \times 5 = 25)$

Answer **all** Questions.

11 a. Construct the truth table for the formula

$$(P \wedge Q) \vee (\neg P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$$

(*Or*)

b. Show the implication.

$$\neg (P \wedge Q) \rightarrow (\neg P \vee (\neg P \vee Q)) \Rightarrow \neg P \vee Q.$$

12 a. Obtain a disjunctive normal form of $\neg (P \vee Q) \leftrightarrow (P \wedge Q)$.

(*Or*)

b. Which of the following propositions are equivalent over the universe of the integers.

i) $0 < n^2 \leq 4$

ii) $0 < n^3 \leq 8$

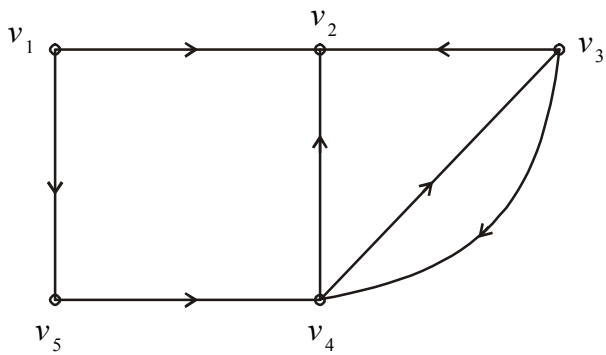
iii) $0 < n \leq 2$

13 a. Let G be a simple graph with n-vertices. Show that if

$\delta(G) \geq \left[\frac{n}{2}\right]$ then G is connected

(*Or*)

b. Find the adjacency matrix, incidence matrix of the following digraph.



14 a. A firm manufactures headache pills in two sizes A and B. Size A contains 2 grains of aspirin, 5 grains of bicarbonate and 1 grain of codeine. Size B contains 1 grain aspirin, 8 grains of bicarbonate and 6 grains of codeine. It is found by users that it requires atleast 12 grains of aspirin, 74 grains of bicarbonate and 24 grains of codeine for providing immediate effect. It is required to determine the least number of pills *a* patient should take to get immediate relief. Formulatism problem as a standard LPP.

(*Or*)

b. Solve graphically the following LPP.

Maximize $Z = 7x_1 + 3x_2$ Subject to the constraints :

$x_1 + 2x_2 \geq 3$, $x_1 + x_2 \leq 4$, $0 \leq x_1 \leq 5/2$, $0 \leq x_2 \leq 3/2$.

15 a. Consider the following transportation problem :

Source	Destination				Availability
	1	2	3	4	
1	20	22	17	4	120
2	24	37	9	7	70
3	32	37	20	15	50
Requirement	60	40	30	110	

(*Or*)

b. How would you deal with the assignment problem where the objective function is to be maximized ? Describe the steps involved in solving unbalanced Assignment problem.

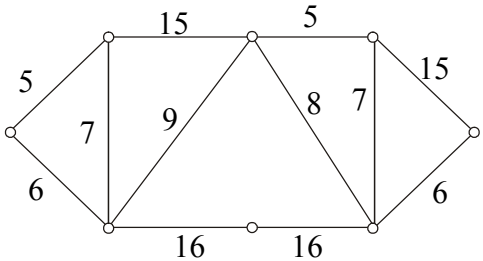
Part - C $(3 \times 10 = 30)$

Answer any **Three** questions.

16. Show that $(P \wedge Q) \rightarrow R \Leftrightarrow (P \rightarrow R) \vee (Q \rightarrow R)$.

17. Prove that $(\exists x)\,(P(x) \wedge Q(x)) \Rightarrow (\exists x)\,P(x) \wedge (\exists x)\,Q(x)$.

18. Find the shortest spanning tree using Kruskal's algorithm.



19. Use two-phase simplex method to maximize $Z = 5x_1 + 8x_2$. Subject to the contraints $3x_1 + 2x_2 \geq 3$, $x_1 + 4x_2 \geq 4$, $x_1 + x_2 \leq 5$, $n_1, n_2 \geq 0$.

20. Solve the following transportation problem.

		To		
From	A	B	C	Available
I	6	8	4	14
II	4	9	8	12
III	1	2	6	5
Demand	6	10	15	

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